

Replication Plan

SCALE Depletion Capabilities for Molten Salt Reactors and Other Liquid-Fueled Systems

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Auto-generated Replication Blueprint

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Paper Overview

This paper describes extensions to the SCALE code system (specifically TRITON and ORIGEN modules) for simulating fuel depletion in molten salt reactors (MSRs) and other liquid-fueled nuclear systems. Key features include modeling continuous material feeds and removals (on-line reprocessing), flowing fuel with delayed neutron precursor transport, and salt treatment processes. The paper validates the methodology against the Molten Salt Reactor Experiment (MSRE) and presents test cases for various MSR concepts.

Detailed Workflow

Phase 1: SCALE Installation and Setup

- 1.1 Obtain and install SCALE 6.3+ (requires ORNL/RSICC license).
- 1.2 Verify nuclear data libraries are installed (ENDF/B-VII.1 or VIII.0).
- 1.3 Test installation with standard verification problems.

Phase 2: Reactor Model Construction

- 2.1 Define the reactor geometry in SCALE input format.
 - MSRE-like single-fluid reactor with graphite moderator.
 - Salt composition: $\text{LiF-BeF}_2\text{-ZrF}_4\text{-UF}_4$ (as specified).
- 2.2 Define material compositions.
 - Fuel salt isotopic composition (enrichment, carrier salt).
 - Moderator and structural materials.

Phase 3: Depletion with Material Flows

- 3.1 Set up TRITON depletion sequence with the liquid-fuel extensions.
- 3.2 Define continuous feed/removal streams.
 - Fission gas removal (Kr, Xe) with specified removal rates.
 - Noble metal removal.
 - Fuel salt feed (makeup uranium or thorium).
 - Protactinium removal (for Th-U cycle).
- 3.3 Define depletion parameters.
 - Power level, irradiation time steps, predictor-corrector settings.
- 3.4 Run the depletion calculation.

Phase 4: Post-Processing and Analysis

- 4.1 Extract nuclide inventories vs. time.
- 4.2 Compute k_{eff} vs. burnup.
- 4.3 Track fissile inventory (U-233, U-235, Pu-239).
- 4.4 Compute neutron spectra at selected time points.
- 4.5 Compute waste stream compositions.

Phase 5: MSRE Validation

- 5.1 Reproduce MSRE benchmark results.
- 5.2 Compare k_{eff} and isotopic compositions against historical MSRE data.
- 5.3 Reproduce figures and tables from the paper.

Datasets

Dataset / Source	Type	Location / Notes
ENDF/B nuclear data	Cross-section libraries	li- Distributed with SCALE; also https://www.nndc.bnl.gov/endl/
MSRE operational data	Historical experimental	ORNL reports: ORNL-4449, ORNL-4396 https://www.osti.gov/
SCALE example inputs	Bundled	Distributed with SCALE package
Paper's input parameters	Specified	Geometry, compositions, flow rates from paper tables

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
SCALE	6.3+	Nuclear analysis code system (TRITON, ORIGEN, KENO) https://www.ornl.gov/scale Licensed via RSICC: https://rsicc.ornl.gov/
OpenMC	latest	Open-source alternative for neutronics https://openmc.org/
Serpent 2	latest	Alternative Monte Carlo depletion code http://montecarlo.vtt.fi/
Python	≥ 3.8	Post-processing and plotting
Matplotlib	≥ 3.4	Visualization of depletion curves

Hardware Requirements

- Multi-core workstation or HPC cluster.
- KENO Monte Carlo: benefits from many cores (≥ 16).
- ≥ 32 GB RAM for detailed depletion with many nuclides.